

Concept of Variation

When two or more quantities depend each other and if one of them changes other item also changes, this is called **variation**.

$$A \propto \frac{1}{B}$$

$$\boxed{A \propto B}$$

AT BT

$$\triangle \frac{AT}{B \downarrow}$$

Concept of Variation

milk price ↑↑



Concept of Variation



milk price $\propto$
↑

milk products price
↑

Concept of Variation

11:07:42



11:00:11

$$S \propto \frac{1}{t}$$

$$S \uparrow$$
$$S = \frac{d}{t}$$

Concept of Variation

When two or more quantities depend each other and if one of them changes other item also changes, this is called variation. For example, if the price of milk increases then cost of related Milk products also increases, that is, the cost of ghee and sweet also increases, this is called as **direct variation**, another example of direct variation can be, if the salary of a person increases, then his Savings and expenditure also increases. On the contrary if speed of a car increases then time to reach destination will decrease or if speed of a car decreases then time to reach destination will increase, this is called as **indirect variation**.

Concept of Variation

Direct Variation

If A increases then B also increases or if A decreases then B also decreases that is A is directly proportional to B, this is called as direct variation. A directly varies to B.

Indirect Variation

If A increases then B decreases or if A decreases then B increases that is A is indirectly or inversely proportional to B, this is called as indirect variation. A indirectly varies to B.



Concept of Variation

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$$\begin{aligned} A &\propto B \\ \boxed{A} &= k \boxed{B} \\ A \uparrow &\quad k = \text{const} \\ \frac{A}{B} &= k \\ \frac{A}{B} &= \text{const} \\ \frac{A_1}{B_1} &= \frac{A_2}{B_2} = \frac{A_3}{B_3} \dots \end{aligned}$$

Concept of Variation

Direct Variation

If A increases then B also increases or if A decreases then B also decreases that is A is directly proportional to B, this is called as direct variation. A directly varies to B.

ratio $\nearrow$ RD $\searrow$ direct

$$A \propto \frac{1}{B}$$

$$A = \frac{K}{B}$$

$$AB = K$$

$$AB = \text{const}$$

$$A_1 B_1 = A_2 B_2 = A_3 B_3 = \dots$$

Indirect Variation

If A increases then B decreases or if A decreases then B increases that is A is indirectly or inversely proportional to B, this is called as indirect variation. A indirectly varies to B.



Que: A varies directly as B. A is 12 when B is 6, what is the value of A when B is 12?



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Que: A varies directly as B. A is 12 when B is 6, what is the value of A when B is 12?

$$A \propto B$$
$$A = k B$$

$$12 = k(6)$$

$$k = 2$$

$$A = k B$$

$$A = 2B$$

$$A = 2 \times 12$$
$$A = 24$$

$$A \propto B \rightarrow$$

Ratio

$$\frac{A_1}{B_1} = \frac{A_2}{B_2}$$

$$\frac{12}{6} = \frac{A_2}{12}$$

$$A_2 = 24$$

Que: A varies directly as B. A is 12 when B is 6, what is the value of A when B is 12?

Handwritten solution showing the relationship between A and B:

$$\begin{array}{ccc} A & \xrightarrow{6} & B \\ 12 & \div 2 & \textcircled{6} \end{array}$$

From the above, $6 \times 2 = 12$, so $12 \times 2 = 24$.

$$\begin{array}{ccc} 12 & \div 2 & \textcircled{6} \\ \times 2 & & \times 2 \end{array}$$

Therefore, $12 \times 2 = 24$.

The final answer is 24.

Que: Y is inversely proportional to X and that $Y=0.4$ when $X=5$. Find Y when $X=4$?



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Que: Y is inversely proportional to X and that $Y=0.4$ when $X=5$. Find Y when $X=4$?

$$Y \propto \frac{1}{X} \quad (IP)$$

$$XY = \text{const}$$

$$Y = \frac{k}{X}$$

$$Y = \frac{k}{X}$$

$$0.4 = \frac{k}{5}$$

$$k = 0.4 \times 5 = 2$$

$$Y = \frac{2}{X}$$

$$Y = \frac{2}{4} = \frac{1}{2}$$

$$= 0.5$$

$$X_1 Y_1 = X_2 Y_2$$

$$5(0.4) = 4(Y_2)$$

$$5 \times \frac{4}{10} = 4(Y_2)$$

$$Y_2 = \frac{5}{10} = \underline{0.5} \quad \checkmark$$